

The Relational Model

Lecture 02

Borough of Manhattan Community College

Introduction

- ❑ A database schema is written using a Data Definition Language (DDL). Unfortunately, this type of language is too low level to describe the data requirements of an organization in a way that is readily understandable by a variety of users.
- ❑ What we require is a higher-level description of the schema: that is, a **data model**. Data model is an integrated collection of concepts for describing and manipulating data, relationships between data, and constraints on the data in an organization.
- ❑ Examples of data models:
 - The Relational Model
 - Entity-Relationship (ER) Model
 - Enhanced Entity-Relationship (EER) Model

History of the Relational Model

- ❑ The Relational Database Management System (RDBMS) has become the dominant data-processing software in use today.
- ❑ This software represents the second generation of DBMSs and is based on the relational data model proposed by E. F. Codd in 1970.
- ❑ Commercial systems based on the relational model started to appear in the late 1970s and early 1980s. Now there are several hundred RDBMSs for both mainframe and PC environments.

Database Relations

- ❑ The relational model is based on the mathematical concept of a **relation**, which is physically represented as a **table**. Codd, a trained mathematician, used terminology taken from mathematics, principally the *Set Theory* and the *Predicate Logic*.
- ❑ A **relation** is a table with columns and rows.
- ❑ An **attribute** is a named column of a relation.
- ❑ A relation is represented as a two-dimensional table in which the rows of the table correspond to individual records and the table columns correspond to attributes.
- ❑ Attributes can appear in any order and the relation will still be the same relation, and therefore will convey the same meaning.

Database Relations

Attributes

Branch

Relation

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Figure 4.1

Instances of the Branch and Staff relations.

Staff

Relation

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Database Relations

- ❑ A **domain** is the set of allowable values for one or more attributes.
- ❑ Every attribute in a relation is defined on a domain.

Attribute	Domain Name	Meaning	Domain Definition
branchNo	BranchNumbers	The set of all possible branch numbers	character: size 4, range B001–B999
street	StreetNames	The set of all street names in Britain	character: size 25
city	CityNames	The set of all city names in Britain	character: size 15
postcode	Postcodes	The set of all postcodes in Britain	character: size 8
sex	Sex	The sex of a person	character: size 1, value M or F
DOB	DatesOfBirth	Possible values of staff birth dates	date, range from 1-Jan-20, format dd-mmm-yy
salary	Salaries	Possible values of staff salaries	monetary: 7 digits, range 6000.00–40000.00

Database Relations

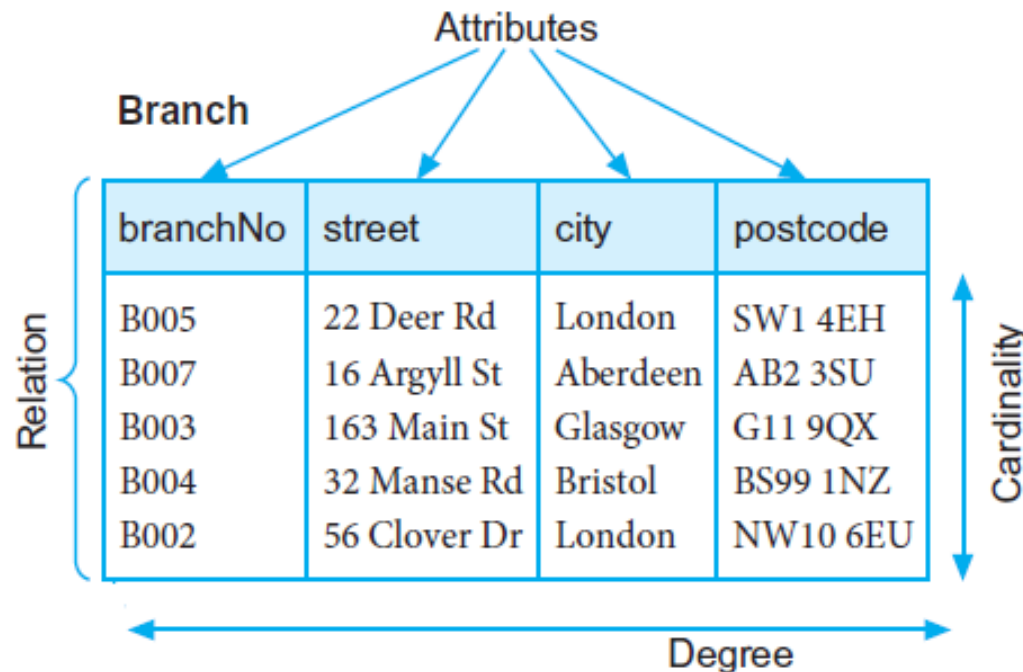
- ❑ A **tuple** is a row of a relation.
- ❑ The elements of a relation are the rows or tuples in the table.
- ❑ Tuples can appear in any order and the relation will still be the same and therefore convey the same meaning.
- ❑ The structure of a relation, together with a specification of the domains and any other restrictions on possible values, is sometimes called its **intension**, which is usually fixed, unless the meaning of a relation is changed to remove or add some attributes.
- ❑ The tuples are called the **extension** of a relation, which changes over time.

Database Relations

- ❑ The **degree** of a relation is the number of attributes it contains.
- ❑ A relation of degree n means that each row of the table contains n values.
- ❑ A relation with only one attribute would have degree one and be called a ***unary*** relation or *one-tuple*.
- ❑ A relation with two attributes is called ***binary***.
- ❑ A relation with three attributes is called ***ternary***, and after that the term *n-ary* is usually used.
- ❑ The degree of a relation is a property of the intension of the relation.

Database Relations

- ❑ The **cardinality** of a relation is the number of tuples it contains.



- ❑ The cardinality is a property of the extension of the relation.

Database Relations

- ❑ A **relational database** is a collection of normalized (appropriately structured) relations with distinct relation names.
- ❑ Alternative terminology:

Alternative terminology for relational model terms

FORMAL TERMS	ALTERNATIVE 1	ALTERNATIVE 2
Relation	Table	File
Tuple	Row	Record
Attribute	Column	Field

Properties of Relations

A relation has the following properties:

- ❑ The relation has a name that is distinct from all other relation names in the relational schema;
- ❑ Each cell of the relation contains at most one atomic (single) value;
- ❑ Each attribute has a distinct name within the relation;
- ❑ The values of an attribute are all from the same domain;
- ❑ Each tuple is distinct; there are no duplicate tuples;
- ❑ The order of attributes has no significance;
- ❑ The order of tuples has no significance.

Relational Keys

- ❑ A **superkey** is an attribute, or set of attributes, that uniquely identifies each tuple within a relation.
- ❑ However, a superkey may contain additional attributes that are not necessary for unique identification.
- ❑ A **candidate key** is a superkey such that no proper subset is a superkey within the relation.
- ❑ Therefore, a candidate key K for a relation R has two properties:
 - Uniqueness: In each tuple of R , the values of K uniquely identify that tuple.
 - Irreducibility: No proper subset of K has the uniqueness property.

Relational Keys

- ❑ Identifying a candidate key requires that we know the “real-world” meaning of the attribute(s) involved so that we can decide whether duplicates are possible.
- ❑ Note that an instance of a relation cannot be used to prove that an attribute or combination of attributes is a candidate key.
- ❑ However, the presence of duplicates in an instance can be used to show that some attribute combination is not a candidate key.

Relational Keys

- ❑ There may be several candidate keys for a relation.
- ❑ A **primary key** is the candidate key that is selected to identify tuples uniquely within the relation.
- ❑ The candidate keys that are not selected to be the **primary key** are called **alternate keys**.
- ❑ A **foreign key** is an attribute, or set of attributes, within one relation that matches the candidate key of **another** relation.
- ❑ When an attribute appears in more than one relation, its appearance usually represents a relationship between tuples of the two relations.

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Primary key

Foreign key

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

PropertyForRent Relation

PropID	Address	Apt	City	Zip	Type	Price	OwnerID
P01	125 Hudson St	7C	New York	10013	Apartment	\$2,000	O61
P02	250 Broadway	1A	Bronx	10718	Apartment	\$1,800	O79
P03	125 Hudson St	3F	New York	10013	Apartment	\$1,700	O61
P04	8806 Third Av		Bay Ridge	11209	House	\$2,500	O14
P05	250 Broadway	1A	Jersey City	07306	Apartment	\$1,900	O53

❑ Foreign key:

- OwnerID

❑ Candidate keys:

- PropID
- Address, Apt, City
- Address, Apt, Zip

❑ Primary key:

- PropID

❑ Alternate keys:

- Address, Apt, City
- Address, Apt, Zip

Enrollment Relation

EnrollmentID	StudentID	Name	CourseID	CourseTitle	Semester	Grade
E05	S021	John White	CIS 395	DB Systems	Fall 2021	D
E07	S021	John White	CIS 395	DB Systems	Spring 2022	A
E11	S038	John White	CIS 395	DB Systems	Spring 2022	C
E18	S046	Alisa Garcia	ACC 215	Statistics	Spring 2022	A
E19	S046	Alisa Garcia	MAT 455	Statistics	Spring 2022	B

❑ Foreign key:

- StudentID
- CourseID

❑ Candidate keys:

- EnrollmentID
- StudentID, CourseID, Semester
- ~~➤ Name, CourseID, Semester~~
- ~~➤ StudentID, CourseTitle, Semester~~

❑ Primary key:

- EnrollmentID

❑ Alternate keys:

- StudentID, CourseID, Semester

Relational Keys - Exercise

The following tables form part of a database held in a relational DBMS:

- **Hotel** (hotelNo, hotelName, city)
- **Room** (roomNo, hotelNo, type, price)
- **Guest** (guestNo, guestName, guestAddress, guestEmail)
- **Booking** (guestNo, roomNo, hotelNo, dateFrom, dateTo)

Assumption: A hotel name cannot be repeated in the same city.

□ Identify, for each table, the followings:

- Foreign keys
- Candidate keys
- Primary key
- Alternate keys

Hotel

hotelNo	hotelName	city
H01	Hilton	New York
H04	Hilton	Boston
H07	Mariott	Washington

⋮

Room

roomNo	hotelNo	type	price
101	H01	Double	140
203	H01	Single	95
101	H07	Double	130

⋮

Guest

guestNo	guestName	guestAddress	guestEmail
G100	David Lee	123 5th Av	Ldavid@gmail.com
G200	Anna Bolt	19 Main St	Anna66@aol.com

⋮

Booking

guestNo	roomNo	hotelNo	dateFrom	dateTo
G100	101	H01	9/2/2020	9/9/2020
G200	101	H07	12/21/2020	12/26/2020

⋮

Assumption:

A hotel name cannot be repeated in the same city.

Identify for each table, the followings:

- Foreign keys
- Candidate keys
- Primary key
- Alternate keys

Relational Keys - Exercise

- ❑ **Foreign keys:**

- ✓ None

- ❑ **Candidate keys:**

- ✓ hotelNo
- ✓ hotelName, city

- ❑ **Primary key:**

- ✓ hotelNo

- ❑ **Alternate keys:**

- ✓ hotelName, city

Hotel

hotelNo	hotelName	city
H01	Hilton	New York
H04	Hilton	Boston
H07	Mariott	Washington
⋮		

Assumption:

A hotel name cannot be repeated in the same city.

Relational Keys - Exercise

- ❑ **Foreign keys:**

- ✓ hotelNo

- ❑ **Candidate keys:**

- ✓ roomNo, hotelNo

- ❑ **Primary key:**

- ✓ roomNo, hotelNo

- ❑ **Alternate keys:**

- ✓ None

Room

roomNo	hotelNo	type	price
101	H01	Double	140
203	H01	Single	95
101	H07	Double	130
⋮			

Relational Keys - Exercise

☐ Foreign keys:

- ✓ None

☐ Candidate keys:

- ✓ guestNo
- ✓ guestEmail
- ✓ guestName, guestAddress

☐ Primary key:

- ✓ guestNo

☐ Alternate keys:

- ✓ guestEmail
- ✓ guestName, guestAddress

Guest

guestNo	guestName	guestAddress	guestEmail
G100	David Lee	123 5th Av	Ldavid@gmail.com
G200	Anna Bolt	19 Main St	Anna66@aol.com

⋮

Relational Keys - Exercise

❑ Foreign keys:

- ✓ guestNo
- ✓ hotelNo
- ✓ roomNo, hotelNo

Booking

guestNo	roomNo	hotelNo	dateFrom	dateTo
G100	101	H01	9/2/2020	9/9/2020
G200	101	H07	12/21/2020	12/26/2020
G100	101	H07	12/21/2020	12/23/2020

⋮

❑ Candidate keys:

- ✓ roomNo, hotelNo, dateFrom
- ✓ roomNo, hotelNo, dateTo

Primary key of Room table

❑ Primary key:

- ✓ roomNo, hotelNo, dateFrom

❑ Alternate keys:

- ✓ roomNo, hotelNo, dateTo

Instance of the DreamHome Rental Database

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Instance of the DreamHome Rental Database

Client

clientNo	fName	lName	telNo	prefType	maxRent	eMail
CR76	John	Kay	0207-774-5632	Flat	425	john.kay@gmail.com
CR56	Aline	Stewart	0141-848-1825	Flat	350	astewart@hotmail.com
CR74	Mike	Ritchie	01475-392178	House	750	mritchie01@yahoo.co.uk
CR62	Mary	Tregear	01224-196720	Flat	600	maryt@hotmail.co.uk

PrivateOwner

ownerNo	fName	lName	address	telNo	eMail
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212	jkeogh@lhh.com
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419	cfarrel@gmail.com
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728	tinam@hotmail.com
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025	tony.shaw@ark.com

Registration

clientNo	branchNo	staffNo	dateJoined
CR76	B005	SL41	2-Jan-13
CR56	B003	SG37	11-Apr-12
CR74	B003	SG37	16-Nov-11
CR62	B007	SA9	7-Mar-12

Viewing

clientNo	propertyNo	viewDate	comment
CR56	PA14	24-May-13	too small
CR76	PG4	20-Apr-13	too remote
CR56	PG4	26-May-13	
CR62	PA14	14-May-13	no dining room
CR56	PG36	28-Apr-13	

Representing Relational DB Schemas

□ A relational database consists of any number of normalized relations. The relational schema for the *DreamHome* case study is:

- **Branch** (branchNo, street, city, postcode)
- **Staff** (staffNo, fName, IName, position, sex, DOB, salary, branchNo)
- **PropertyForRent** (propertyNo, street, city, postcode, type, rooms, rent, ownerNo, staffNo, branchNo)
- **Client** (clientNo, fName, IName, telNo, prefType, maxRent, eMail)
- **PrivateOwner** (ownerNo, fName, IName, address, telNo, eMail, password)
- **Registration** (clientNo, branchNo, staffNo, dateJoined)
- **Viewing** (clientNo, propertyNo, viewDate, comment)

Integrity Constraints

- A data model has three parts: **structural part**, discussed in the previous section, a **manipulative part**, defining the types of operation that are allowed on the data, and a set of **integrity constraints**, which ensure that the data is accurate.
- 1. **Domain constraints** are an example of integrity constraints. They are restrictions on the set of values allowed for the attributes.
- 2. **General constraints:** Additional rules specified by the users or DBAs that define or constrain some aspect of the enterprise.

Course Load Limit Example:

Students cannot take more than 18 credits per semester.

Integrity Constraints

- ***Null*** represents a value for an attribute that is currently unknown or is not applicable for this tuple. *Null* is not a value but represents the absence of a value.
3. **Entity integrity:** In a base relation, no attribute of a primary key can be *null*.

If we allow a *null* for any part of a primary key, we are implying that not all the attributes are needed to distinguish between tuples, which contradicts the definition of the candidate key.

4. **Referential integrity:** If a foreign key exists in a relation, either the foreign key value must match a candidate key value of some tuple in its home relation or the foreign key value must be wholly ***null***.

Integrity Constraints

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Primary key

Foreign key

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Views

- ❑ A **base relation** is a named relation whose tuples are physically stored in the database.
- ❑ A **view** is the dynamic result of one or more relational operations operating on the base relations to produce another relation.
- ❑ Views are **dynamic**, meaning that changes made to the base relations that affect the view are immediately reflected in the view. When users make permitted changes to the views, these changes are made to the underlying relations.

Examples of Views

- ❑ Some members of staff should see **Staff** tuples without the **salary** attribute.

Base relation

Staff					
staffNo	name	position	sex	salary	branchNo
SL21	John White	Manager	M	30000	B005
SG37	Ann Beech	Assistant	F	12000	B003
SG14	David Ford	Manager	M	26000	B007
SA9	Mary Howe	Assistant	F	9000	B007
SG5	Susan Bran	Manager	F	24000	B003
SL41	Julie Lee	Assistant	F	9000	B005

View

Staff2				
staffNo	name	position	sex	branchNo
SL21	John White	Manager	M	B005
SG37	Ann Beech	Assistant	F	B003
SG14	David Ford	Manager	M	B007
SA9	Mary Howe	Assistant	F	B007
SG5	Susan Bran	Manager	F	B003
SL41	Julie Lee	Assistant	F	B005

Examples of Views

- Attributes may be renamed or the order of attributes changed. Records can be filtered.

Base relation

Staff

staffNo	name	position	sex	salary	branchNo
SL21	John White	Manager	M	30000	B005
SG37	Ann Beech	Assistant	F	12000	B003
SG14	David Ford	Manager	M	26000	B007
SA9	Mary Howe	Assistant	F	9000	B007
SG5	Susan Bran	Manager	F	24000	B003
SL41	Julie Lee	Assistant	F	9000	B005

View

Staff3

staffNo	branchNo	managerName	position	sex	yearlySalary
SL21	B005	John White	Manager	M	30000
SG14	B007	David Ford	Manager	M	26000
SG5	B003	Susan Bran	Manager	F	24000

Examples of Views

- A user might need **Branch** tuples that contain the names of managers as well as the other attributes already in **Branch**. This view is created by combining the **Branch** relation with the **Staff** relation where the staff position is “*Manager*”.

Branch

branchNo	street	city	postcode
B003	163 Main St	Glasgow	G11 9QX
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU

Staff

staffNo	name	position	sex	salary	branchNo
SL21	John White	Manager	M	30000	B005
SG37	Ann Beech	Assistant	F	12000	B003
SG14	David Ford	Manager	M	26000	B007
SA9	Mary Howe	Assistant	F	9000	B007
SG5	Susan Bran	Manager	F	24000	B003
SL41	Julie Lee	Assistant	F	9000	B005

Base relations

BranchManagers

branchNo	street	city	postcode	name
B003	163 Main St	Glasgow	G11 9QX	Susan Bran
B005	22 Deer Rd	London	SW1 4EH	John White
B007	16 Argyll St	Aberdeen	AB2 3SU	David Ford

View

Purpose of Views

- ❑ The view mechanism is desirable for several reasons:
 - It provides a powerful and flexible security mechanism by hiding parts of the database from certain users. Users are not aware of the existence of any attributes or tuples that are missing from the view.
 - It permits users to access data in a way that is customized to their needs, so that the same data can be seen by different users in different ways, at the same time.
 - It can simplify complex operations on the base relations. For example, if a view is defined as a combination (join) of two relations, users may now perform more simple operations on the view, which will be translated by the DBMS into equivalent operations on the base relations.

Updating Views

- ❑ All updates to a base relation should be immediately reflected in all views that reference that base relation.
- ❑ Similarly, if a view is updated, then the underlying base relation should reflect the change. However, there are restrictions on the types of modification that can be made through views:
 - Updates are allowed through a view defined using a simple query involving a single base relation and containing either the primary key or a candidate key of the base relation. Thus, updates are not allowed through views involving multiple base relations.
 - Updates are not allowed through views involving aggregation or grouping operations.

Updating Views

Why updates are not allowed through views involving multiple base relations?

Staff

staffNo	name	position	sex	salary	branchNo
SL21	John White	Manager	M	30000	B005
SG37	Ann Beech	Assistant	F	12000	B003
SG14	David Ford	Manager	M	26000	B007
SA9	Mary Howe	Assistant	F	9000	B007
SG5	Susan Bran	Manager	F	24000	B003
SL41	Julie Lee	Assistant	F	9000	B005

Branch

branchNo	street	city	postcode
B003	163 Main St	Glasgow	G11 9QX
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU

View

Staff4

staffNo	name	position	sex	salary	city
SL21	John White	Manager	M	30000	London
SG14	David Ford	Manager	M	26000	Aberdeen
SG5	Susan Bran	Manager	F	24000	Glasgow

If updates are allowed, how do we interpret the change of the city of the first record in the view from London to Glasgow?

Updating Views

Why updates are not allowed through views involving aggregation or grouping operations?

Staff

staffNo	name	position	sex	salary	branchNo
SL21	John White	Manager	M	30000	B005
SG37	Ann Beech	Assistant	F	12000	B003
SG14	David Ford	Supervisor	M	18000	B003
SA9	Mary Howe	Assistant	F	9000	B007
SG5	Susan Bran	Manager	F	24000	B003
SL41	Julie Lee	Assistant	F	9000	B005

StaffSalaries

branchNo	count	sumSalaries
B003	3	54000
B005	2	39000
B007	1	9000

View

If the sum of the salaries in branch B003 increases from 54000 to 60000, how will the extra 6000 be distributed among the three workers of B003?